

Value-Guided Action Planning with JEPA World Models

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Joint-Embedded Predictive Architectures

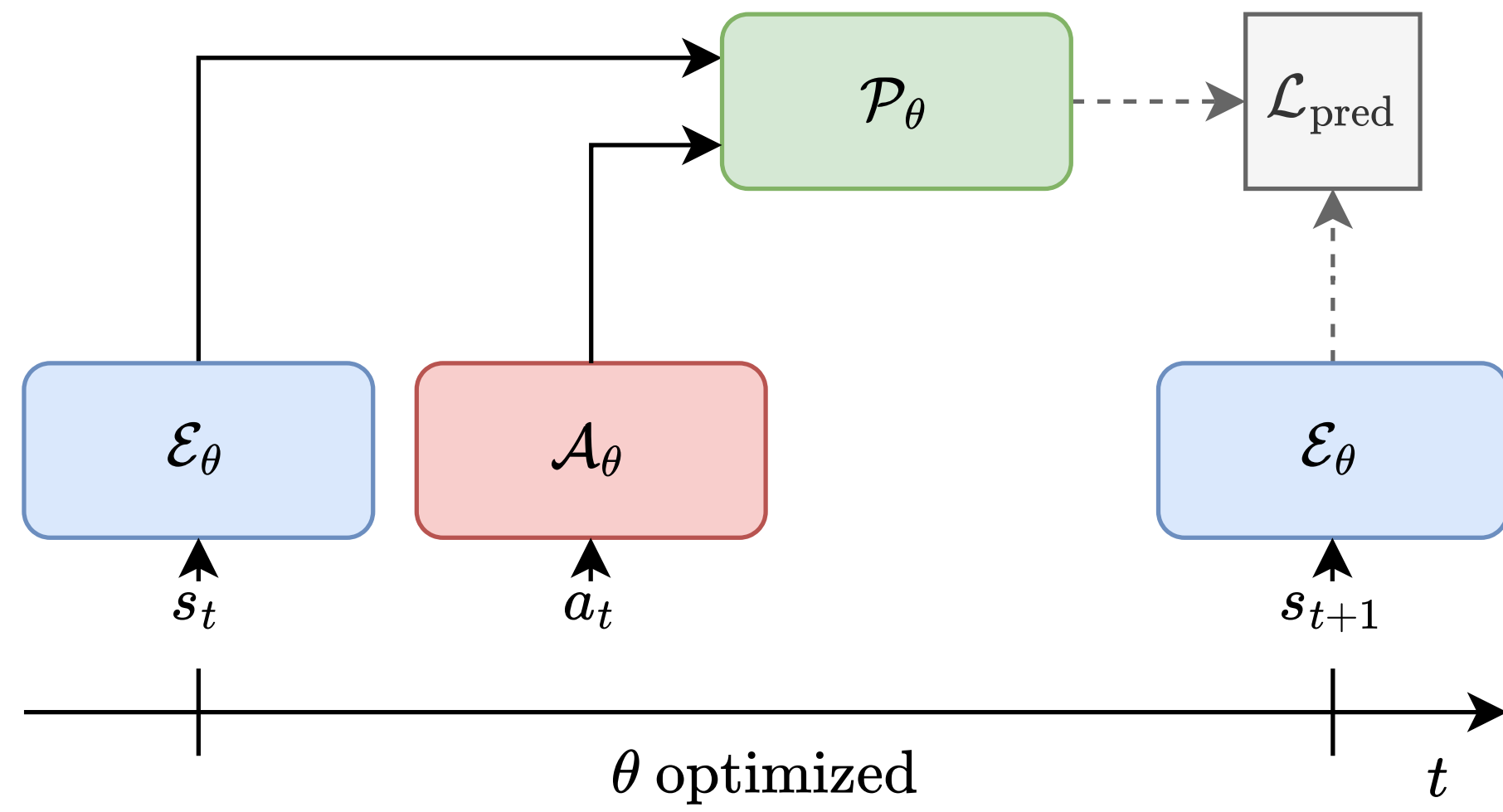


Figure 1. Diagram of a JEPA model at train time

- Trained with trajectories in an environment, using **L2 prediction loss** and **collapse-prevention approaches** (e.g., VCReg loss, EMA).
- Can be used for action planning given a goal and a start state [4]. Current works use **Model Predictive Control (MPC)**, which iteratively minimizes the distance between predicted and goal representations.
- Issues:** This planning approach may encounter local minima and it is limited by a finite horizon.

Learning value-aware representations

- Goal conditioned value functions** between state s and goal g :

$$V^*(s, g) = \lim_{T \rightarrow +\infty} \sup_{(a_t)} \sum_{t=0}^T -\gamma^t C(s_t, a_t, g), \text{ with } \begin{cases} s_0 = s \\ \forall t, s_{t+1} = d(s_t, a_t) \end{cases} \quad (1)$$

where d is the dynamics of the environment, $\gamma \in [0, 1]$ is the discount factor, and $C : (s, a, g) \mapsto \mathbf{1}_{s \neq g}$ is a reaching cost.

- Implicit Q-Learning (IQL)** [2] allows learning a goal-conditioned value function from unlabeled trajectories by leveraging **expectile regression**.

$$\mathcal{L}_{VF}^\theta((s_t), (g_n)) = \sum_{n=0}^N \sum_{t=0}^{T-1} L_\tau^2 \left(-\mathbf{1}_{s_t \neq g_n} + \gamma V_\theta(s_{t+1}, g_n) - V_\theta(s_t, g_n) \right), \quad (2)$$

where $L_\tau^2(x) = |\tau - \mathbf{1}_{x < 0}| x^2$.

Goals are the last state of the training trajectories or randomly sampled from the training batches.

- We learn representations such that

$$-\|\mathcal{E}_\theta(s) - \mathcal{E}_\theta(g)\|_2 = V^*(s, g),$$

as in [3]. The Euclidean distance can also be replaced with a generic **quasi-metric distance**, following [5].

- This then allows MPC to account for **longer time horizons** and stabilizes the procedure by **removing local minima** [1].

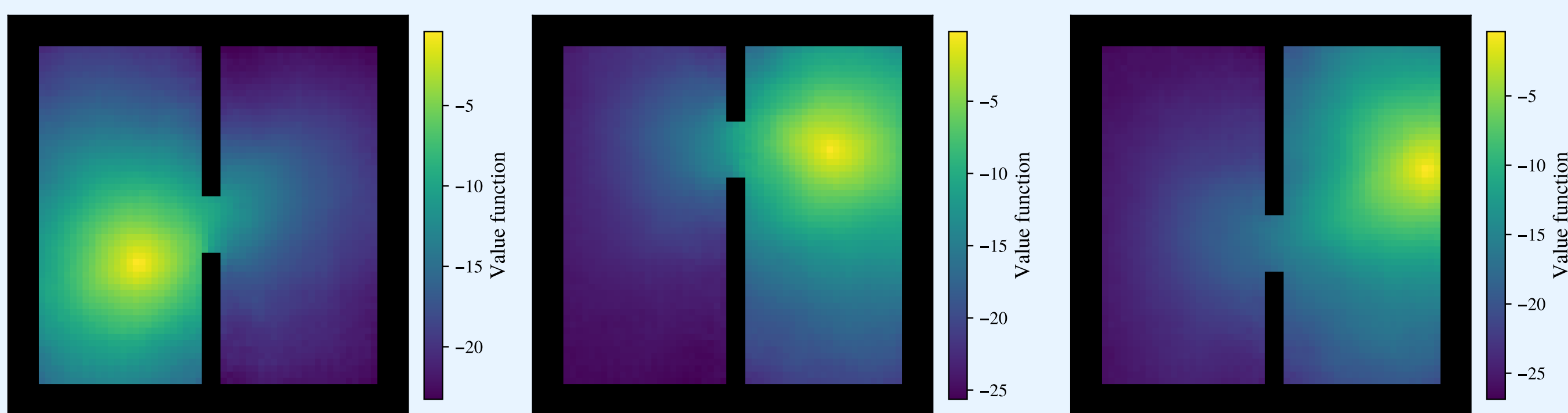


Figure 2. Visualization of $-\|\mathcal{E}_\theta(s) - \mathcal{E}_\theta(g)\|_2$ in a wall environment for three different goals

Baseline alternatives to the IQL loss

- Contrastive loss:**

$$\mathcal{L}_C^\theta((s_t^+), (s_t^-)) = \sum_{t=0}^{T-1} \|\mathcal{E}_\theta(s_{t+1}^+) - \mathcal{E}_\theta(s_t^+)\|_2^2 + (m - \|\mathcal{E}_\theta(s_t^-) - \mathcal{E}_\theta(s_t^+)\|_2)_+, \quad (3)$$

where (s_t^+) is a training trajectory, (s_t^-) a sequence of random states, m a real scalar that can be tuned to control stability, and $x_+ = \max(0, x)$.

- Regressive loss:**

$$\mathcal{L}_R^\theta((s_t)) = \sum_{t=0}^{T-1} (\|\mathcal{E}_\theta(s_{t+1}) - \mathcal{E}_\theta(s_t)\|_2 - 1)^2. \quad (4)$$

Experiment settings

We conduct experiments under an **offline reinforcement learning setting**. Models are trained with random trajectories sampled in the environments.

The **wall environment** is a space with a random wall and door. “WS” uses small-norm actions and “WB” uses large-norm actions.

The **maze environment** features an agent navigating from a random start to a random goal within a random maze, and it accounts for inertia.

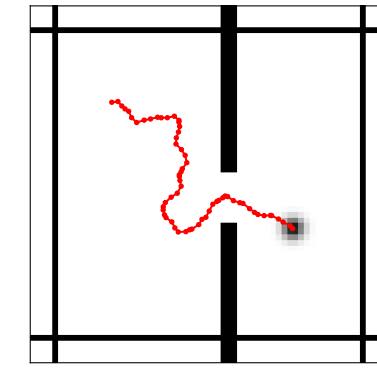


Figure 3. Wall dataset trajectory

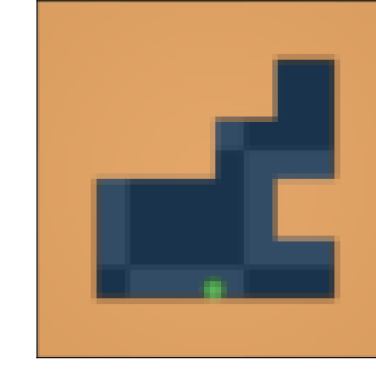


Figure 4. Maze dataset observation

Results

We assess the quality of the learned representations by evaluating the **planning accuracy** of the model (proportion of successful plans for random pairs of initial states and goals).

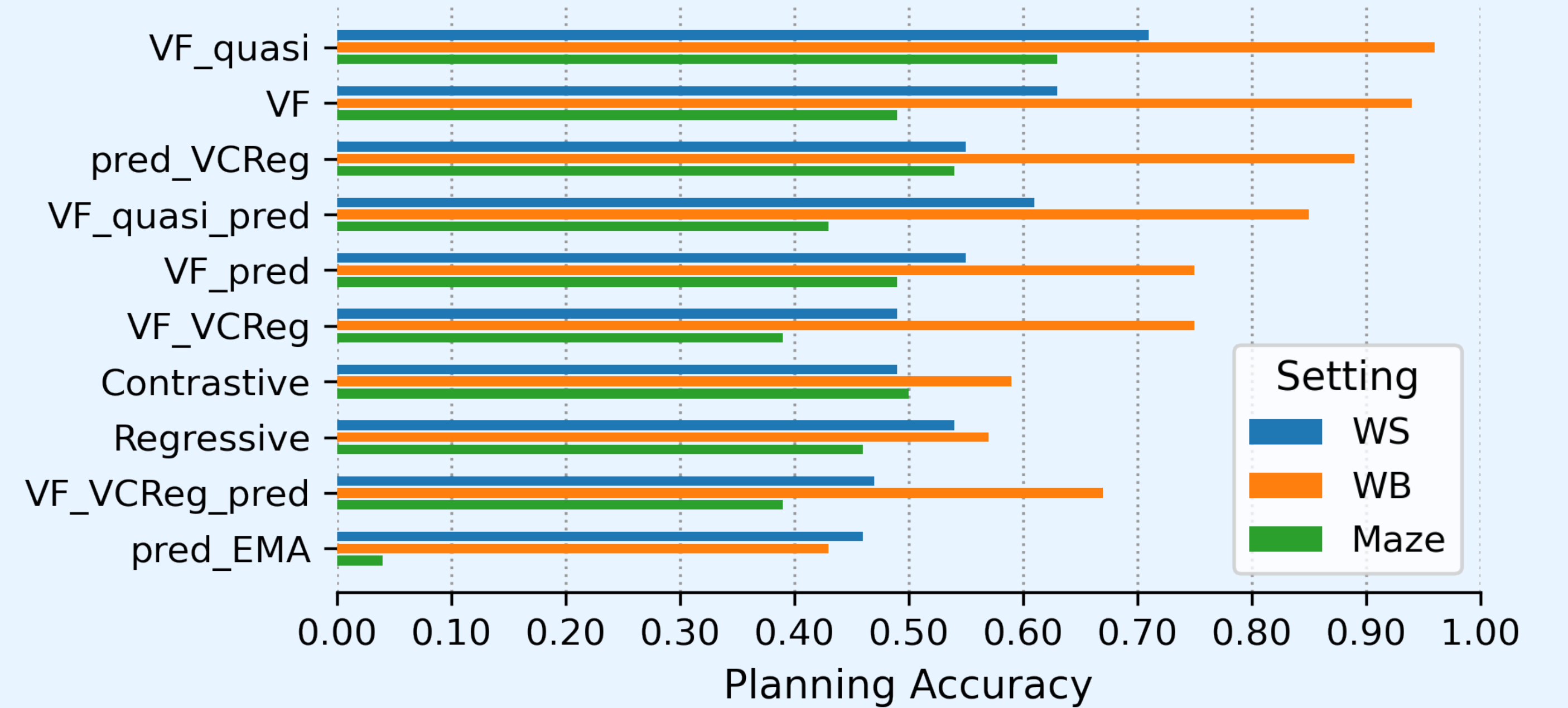


Figure 5. Planning results

“_pred” indicates simultaneous training of the state encoder, the action encoder and the predictor. Otherwise the state encoder is first trained alone with its own loss.

Discussion

- Goal-conditioned value functions provide **limited information for distant states** due to discounting and sparse training examples. Possible solution: **hierarchical approaches**.
- Expert trajectories might be preferred for consistency, but may reduce diversity. **Full state-space coverage remains crucial**, and better-explored datasets improve results.
- Testing the approaches under noise would provide valuable insights, as IQL is known **to be biased in such settings**.
- Even if a stopping criterion cannot be defined for the cost, training can rely on sequences of states to learn an approximated value function.

Takeaways

- Unlike prior JEPA planning work, we learn representations that approximate goal-reaching value functions, **leading to improved planning performance**.
- Unlike prior IQL work, we plan using these representations with a JEPA predictor and MPC, rather than RL methods that learn a critic.
- Evaluating our approach in **random environments** or with **hierarchical architectures** would be valuable.

Bibliography

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